

INFORMATICS INSTITUTE OF TECHNOLOGY

Foundation Certificate in Higher Education

Module: DOC 314 – Digital Skills for Higher Education

Module leader: Ms. Jananie Mayooresan

Assessment type: Group Coursework

Group ID: S-L4

Project Title: Textile Industry (Knitting Technology)

Student details:

Student ID	Name
20221411	M.G.B.Y. Dias
20220745	G.S.M.K. Ganegoda
20221423	H.M.T. Lehan
20221404	T.A. Karunarathne
20221414	A.W.D.D. Abeyrathna
20221424	J.A. Ahamath
20221418	M.S.M. Aflal

Abstract

The textile and apparel industries are significant to the global economy, contributing $1.7 trillion and supporting between 60 and 75 million jobs worldwide. According to reports indicating a strong increase in EU apparel and textile exports during 2014, knitted fabrics made up 7% of exports and 5% of all textiles brought into the EU. Technical textiles can be attributed to a variety of variables, including the expansion of to play, even though this figure may seem shallow in compared to its sister textile production technique (woven). Significant changes within the traditional textile business have been brought about by the remarkable shift in textile markets over the past ten years, with ever-more varied uses pouring into other industries. There is no exception to this market diversification strategy when it comes to some essential applications in the fields of. This industry of textiles has a history of being highly adaptable and resilient to change, with ongoing technological advancement and consistently new applications. If projections come true today, it will look very different from the little cottage industry of the 16th century. In the developed economies, specifically in the area of advanced knitted performance, the knitting industry is currently interesting due to a degree of reshoring (the United States and the United Kingdom) and the rise of African players. Several reports claim that there will be a rapid movement of apparel production, and that the knitwear sector in Morocco (Casablanca) and Ethiopia are also growing in the low-cost sector. The world produces over 17 million tons of knitted textiles and apparel representing one-third of the global textile market. It is clear that one important contributor to this figure is knitted high-performance apparel. It is forecast that knitted textiles and apparel are set to grow by 25% over the next ten years, reaching more than 21 million era, and the knitting sector will not disappoint with novel advanced solutions. Over 17 million tons of knitted textiles and clothing are produced annually, accounting for one-third of the world's total textile output. It is obvious that knitted high-performance clothing is a significant factor in this number. Over the next 10 years, it is predicted that the market for knitted textiles and garments will increase by 25%, reaching more than 21 million people. The knitting industry will not fall short with innovative, cutting-edge solutions.

Acknowledgments

We would like to express our special thank of gratitude to our tutorial lecturer Ms. Chathuri Udagedara as well as the module leader Ms. Jananie Mayooresan for giving us this opportunity to make this report about Knitting technology related to the Textile Industry, which also helped us to learn many new things by doing this research.

Secondly, I would like to thank our colleagues who helped us to finish this report on time.

Table of Content

Abstract i
Acknowledgments ii
Table of Content iii
List of Figures iv
List of Tables iv
1. Introduction 1
2. Importance/role of knitting technology in the Textile Industry 2
2.1 Role of knitting machine 3
3. Detailed analysis of the process of knitting 3
4. Knitting Faults 5
5. Knitted Fabric and their uses 6
6. Who are involved in this process? 7
7. Past, present and future of the knitting technology in the organization 7
7.1 Past 7
7.2 Present 8
7.3 Future 8
8. Conclusion 9
9. References 10
10. Appendices 11
10.1 Workload matrix 11
10.2 Meeting Agenda 12
10.3 Meeting Minutes 13

List of Figures

Figure 1:1.1 Teejay Lanka logo 1
Figure 2:1.2 Teejay Lanka- knitting technology 1
Figure 3:2.1 Knitting machine 2
Figure 4:3.1 Classification of knitting process 3
Figure 5:4.1 Defects in knitted fabric 5
Figure 6:4.2 Knitting faults 5
Figure 7:7.1 Synthetic Fabric 8

List of Tables

Table 1:3.1 Comparison of warp and weft knitting 5

1. Introduction

Since 1986, the textile and apparel sectors have contributed more than 52% of Sri Lanka's overall export revenues, making them the country's top gross export earner. Additionally, it has been the nation's top net earner of foreign currency since 1992. Around half of Sri Lanka's total exports come from the garment sector, which employs about 15% of the nation's workforce and is among the top nations in the world for textile production in terms of population. Among the textile companies in Sri Lanka Teejay Lanka became one of the leading textile-producing companies because of its using knitting technology.

The company was found in 2000 and initially, it was named Textile Jersey Lanka Pvt Ltd. In 2016 it was rebranded as “Teejay” Teejay Lanka. The Teejay Group is positioned as Sri Lanka's flagship and leading multinational manufacturer of weft-knitted fabric. The Teejay Group is driven by a passion to become the preferred fabric solutions provider for branded clothes. The state-of-the-art printing plant and the Strategic Business Unit, which has the same capacity as its HQ created in India but is headquartered in Avissawella, are evidence that Teejay Lanka has expanded its territory to make its presence felt in local and worldwide markets. (Teejay Lanka, 2010)

Figure 1:1.1 Teejay Lanka logo

Teejay Lanka, which employs over 2600 people in Sri Lanka and India, regards fostering innovation and sustainability as well as building relationships as essential elements of its business DNA. Its basic principles of Integrity, Innovate to Lead, Collaboration, and Togetherness serve as the cornerstone on which the company's culture is constructed. (Teejay Lanka, 2010)



Figure 2:1.2 Teejay Lanka- knitting technology

2. Importance/role of knitting technology in the Textile Industry

Since 1000 B.C., knitting technology has been a significant part of our way of life. It produces fabrics and forms garments for numerous technical end applications in addition to clothing and home uses. Modern machinery and extremely complex and specialized textile goods have both benefited much from the knowledge that has been accumulated over the years, especially in the last 50 years in the field of knitting technology. Nowadays, undergraduate, and graduate courses in textile-related fields across the nation contain both the fundamentals of knitting technology and its most recent advancements in their curricula. (Chattopadhyay, 2012)

Once circular knitting machines can produce these fabrics at a high rate and with acceptable quality, they will take the lead in the textile industry in producing special fabrics. This will enable the competitive production of special and industrial fabrics for automotive, medical, and elegant clothing applications. (Semnani, 2011)



Figure 3:2.1 Knitting machine

2.1 Role of knitting machine

Making fabrics (textiles) through the interlacing of strands is the technique of knitting. The two main types of knitting are **weft knitting** and **warp knitting**. Both machines and hands can be used to knit. Making loops of yarn in a row flat or in the round is how knitting is done. You may find a list of the various industrial knitting machine types used to create knitted fabrics in this article. Based on methods used to create fabrics, knitting machines are categorized (warp knitting, weft knitting, single jersey, and double jersey). Machines are categorized according to their configuration. (Online clothing study, 2015)

3. Detailed analysis of the process of knitting

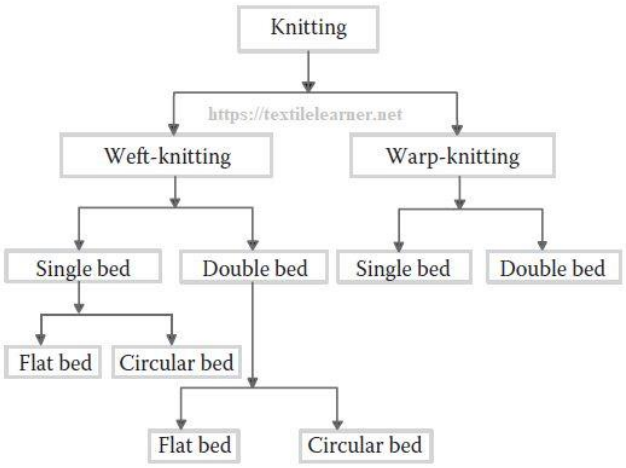


Figure 4:3.1 Classification of knitting process

In Teejay Lanka, Manufacturing is a 3-step process which takes place across three major production units:

1. The Knitting process converts the yarn into Greige.
2. The Dyeing process colors the griege to the specified Color.
3. The Finishing process ensures that the dyed fabric is finished to the exacting standards of our customers.

- Their Quality Department monitors and checks each process to ensure that they maintain our highest standards for our clients. (Teejay Lanka, 2010)

Early flatbed stocking frames had bearded needles made of low carbon steel with reflexed tips that could be pressed into a hollow to close the loop. A needle bar (bed) that moved back and forth, to and from the operator, supported the needles. A presser bar simultaneously depressed the beards.

- The needle bar advances, and the exposed needles dislodge the web.
- The needles are placed with the weft thread.
- Weft thread falls erratically.
- The weft is drawn into the open needles as the needle bar pulls back.
- The presser bar descends, the needle loops shut, the needle bar retracts, and the weft is pulled back through the web.
- The needles open, revealing a fresh row of the web that descends due to gravity.

Knitting comes from the term knot, which itself comes from the Old English verb cnyttan, meaning to tie a knot. Many different styles of clothing employ the knitting technique to manipulate yarn and make textiles or fabrics.

Knitting can be done manually or mechanically. In knitting, loops of yarn are arranged in a row, either flat or round (tubular). On a knitting needle, multiple active stitches are typically present at once. Knitted fabric is made up of several parallel rows of linked loops that touch the rows before and after them. In order to prevent the loops from the previous row from unraveling when the next row is removed off the other needle, each newly generated loop is passed through one or more loops from the row before it is put on the gaining needle.

Knitting is a method for creating a two-dimensional fabric from a one-dimensional yarn or thread, much like weaving. A run or ladder develops when yarn in a conventional knitted fabric break, causing the loops to unravel. Knitters attempted to create new fabric constructions that were less reliant on individual loops for strength and unlikely to run if a thread broke in order to solve the issue and sell more items. Consumers welcomed the introduction of hexagonal meshes, micro-mesh, non-run, sunproof, and similar fabrics.

The differences between the main two knitting processes can be shown as below. (Uzzal, 2019)

Warp knitting process	Weft knitting process
The used yarn is oriented horizontally in this situation.	The used yarn is oriented vertically in this situation.
Compared to weft knitting, there is less shrinking.	Compared to warp knitting, there is more shrinking.
Less elasticity is seen in warp knit fabrics.	More elasticity is seen in weft knit fabrics.
Ideal for producing coarse fabric.	Ideal for producing thin fabric.
Here, each knitting needle needs at least one yarn.	For one yarn, any quantity of knitting needles is required.
It is not difficult to tailor warp knit cloth.	It is difficult to tailor weft knit cloth.

Table 1:3.1 Comparison of warp and weft knitting

4. Knitting Faults

The faults that occur during the process of knitting can be classified as follows. (Uthpala, 2019)

1. Machine setting related
    - Yarn end out
    - Tight barre lines
    - Bowing
    - Oil stains

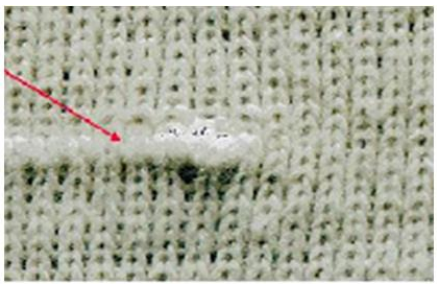


Figure 5:4.1 Defects in knitted fabric

2. Yarn related
    - Contamination
    - Thin yarns
    - GSM variation

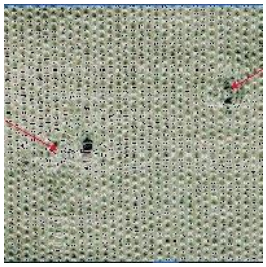


Figure 6:4.2 Knitting faults

3. Knitting elements related
    - Needle lines
    - Broken lines

- Drop stitches
- Sinker lines

5. Knitted Fabric and their uses

Knitting is the second most popular method for making fabric. (2022 percentage of the entire amount of cloth) In the past 30 years, there have been significant developments in knitting for better quality materials and great production output. According to the data, fabric is the main production of textiles by knitting.

A series of loops produced from one or more yarns are interlocked to create knitted fabrics. Additionally manufactured solely by machines, warp knits are typically less elastic, closer, and run-resistant than filler knits. (Textile School, 2018)

Here are a few illustrations of knitted fabric made in Teejay Lanka and their uses.

1. Flat or Jersey Knit Fabric
    - This stitch is used in making nylon hosiery, men's underwear, and t-shirts.

2. Purl knit Fabric
    - It is frequently used to make big sweaters and children's apparel.

3. Rib Stitch Knit Fabric
    - Rib knits are used for the "ribbing" which is usually found at the lower edges of sweaters, on sleeve cuffs, and at necklines.

4. Sliver Knit Fabric
    - They are widely used in making jackets and coats.

5. Intarsia Knit Fabric
    - It is typically used to make shirts, blouses, and sweaters.

6. Knitted Terry Fabric
    - Beachwear, towels, and other items are frequently made of Terry fabric

6. Who are involved in this process?

- Management level
    - Managing the issues.
- Senior executives
    - People who supervise executives.
- Executives
    - Checking machine by machine.
    - Checking designs, loops, etc.
- Team Leaders
    - People who supervise operators.
    - Correcting mistakes done by operators.
- Associates (Operators)
    - People who set up the machines.
    - People who operate the machines.

(Uthpala, 2019)

7. Past, present and future of the knitting technology in the organization

7.1 Past

- In 2000 Teejay Lanka started knitting with few staff members.
- They initially had around 10 knitting machines in their warehouse.
- The operations were done in a single floor with limited space.
- During the early days, only cotton was used in knitting.
- Later in 2010, the number of knitting machines were increased to 50 and the number of workers were increased as well. This helped in increasing the quality of fabric knitted in the organization. (Madusanka, 2022)

7.2 Present

- Currently around 126 different types of knitting machines are used in the organization with a staff around 2600.
- At present they do all types of knitting including single jersey knitting and double jersey knitting.
- They recently added another floor to their organization with 60 more machines.
- Now they use all types of materials including cotton, polyester etc. depending on the customer's needs.
- Currently, all the knitting machines in the organization are connected through a software using Wi-Fi. This increases the efficiency of the knitting process.
- They're currently installing needle scanners to the knitting machines they use. So that they can reduce the cost and time spent on knitting faults. (Madusanka, 2022)

7.3 Future

- In Future they are planning to expand their organization by adding a new building to the Seethavaka export zone.
- They are also planning to add 60 more machines to the new building.
- They have plans to introduce synthetic fabric to their organization in the near future. (Madusanka, 2022)

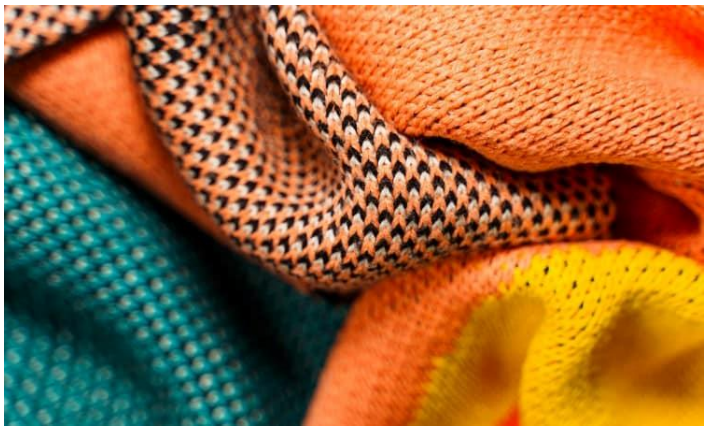


Figure 7:7.1 Synthetic Fabric

8. Conclusion

According to this report, Textile and apparel sectors have contributed more than 52% of Sri Lanka's overall export revenues, making them the country's top gross export earner. Knitting is a method of creating fabrics (textiles) by interlacing threads. Weft knitting and warp knitting are the two primary types of knitting. Knitting may be done by hands or by machines. Knitting is the second most popular method of fabric production. Teejay Lanka became one of the leading textile-producing companies because of it using knitting technology. Knitting technology will get more advanced with the time and different types of fabric will be introduced in the future.

9. References

Chattopadhyay, P. (., 2012. *Knitting Technology - an overview.* [Online]
Available at: https://www.sciencedirect.com/topics/engineering/knitting-technology#:~:text=2%20Knitted%20fabrics-,Knitting%20technology%20produces%20a%20fabric%20with%20meshes%20of%20interlocking%20loops,or%20disadvantages%20for%20electronic%20applications.
[Accessed 18 11 2022].

Madusanka, M. N., 2022. *Present, Past and Future of knitting technology in Teejay Lanka* [Interview] (20 11 2022).

Online clothing study, 2015. *Classification of knitting machines.* [Online]
Available at: https://www.onlineclothingstudy.com/2020/08/classification-of-knitting-machines.html
[Accessed 19 11 2022].

Semnani, D., 2011. *Avances of knitting technology.* [Online]
Available at: https://www.sciencedirect.com/science/article/pii/B9781845693725500073
[Accessed 18 11 2022].

Teejay Lanka, 2010. *Company Overview.* [Online]
Available at: http://www.teejay.com/about-teejay/company-overview
[Accessed 17 11 2022].

Teejay Lanka, 2010. *Company Overview.* [Online]
Available at: http://www.teejay.com/about-teejay/company-overview
[Accessed 17 11 2022].

Teejay Lanka, 2010. *Service ranges.* [Online]
Available at: http://www.teejay.com/our-offers-and-customers/service-ranges
[Accessed 19 11 2022].

Textile School, 2018. *Knitted fabrics and types.* [Online]
Available at: https://www.textileschool.com/251/knitted-fabrics-and-types/
[Accessed 19 11 2022].

Uthpala, V., 2019. Knitting Faults. In: *Lectue Notes.* Colombo: s.n., p. 24.

Uzzal, S. H., 2019. *Comparison between warp and weft knitting textile.* [Online]
Available at: https://garmentsmerchandising.com/warp-weft-knitting-textile-difference/
[Accessed 20 11 2022].

10. Appendices

10.1 Workload matrix

Allocated Task	20221411	20220745	20221423	20221404	20221414	20221424	20221418
Introduction	✓			✓	✓		
Importance/role of knitting technology in the Textile Industry	✓		✓	✓		✓	✓
Detailed analysis of the process of knitting	✓	✓			✓	✓	✓
Knitting faults			✓	✓	✓		
Knitted Fabric and their uses		✓				✓	
Past, present, and future of the knitting technology in the organization	✓	✓	✓				✓

10.2 Meeting Agenda

1st Coursework Meeting

1. Call the meeting.
2. Brief discussion between the team members.
3. Appoint a team leader.
4. Discuss the proceeding of the coursework.
5. Plan the coursework.
6. Agree on a date and time for the next meeting.

2nd Coursework Meeting

1. Call the meeting.
2. Minutes of the last meeting.
3. Implementation of a plan for the report.
4. Set a date and time for the next meeting.

3rd Coursework Meeting

1. Call the meeting.
2. Minutes of the last meeting.
3. Share the report with the team.
4. Discuss the proceedings of the PowerPoint Presentation.
5. Agree on a date and time for the next meeting.

4th Coursework Meeting

1. Call the meeting to order.
2. Minutes of the last meeting.
3. Brief discussion about the PowerPoint Presentation.
4. Join a meeting with some specialist in textile industry.
5. Enter more facts and share the report with the group members.
6. Agree on a date and time for the next meeting.

5th Coursework Meeting

1. Call the meeting.
2. Minutes of the previous presentation.
3. Finalize the report.
4. Finalize the Presentation.
5. Set a deadline for rehearsal.
6. Fix a time for the submission.

10.3 Meeting Minutes

1st Coursework Meeting

Date: 7th of November 2022

Time: 10.30 a.m. – 12.30 p.m.

Venue: Virtual Zoom meeting

Attended by:

1) M. G. B. Y. Dias
2) G. S. M. K. Ganegoda
3) H. M. T. Lehan
4) T. A. Karunarathne
5) A. W. D. D. Abeyrathna
6) J. Asfack Ahamath
7) M. S. M. Aflal

Time	Minutes
10.30 a.m.	The meeting started with introductions by all the team members of group L4.
10.50 a.m.	A discussion was held to appoint the group leader Mr. M. G. B.Y. Dias was appointed as the group leader.
11.00 a.m.	A plan was formulated by all the team members to complete the coursework by the deadline.
11.45 a.m.	The tasks were divided among the group members.
12.30 p.m.	The meeting was adjourned until 9th of November.

2nd Coursework Meeting

Date: 9th of November 2022

Time: 04.00 p.m. – 5.30 p.m.

Venue: Cafeteria, IIT Business School

Attended by:

1) M. G. B. Y. Dias
2) G. S. M. K. Ganegoda
3) H. M. T. Lehan
4) T. A. Karunarathne
5) A. W. D. D. Abeyrathna
6) J. Asfack Ahamath

Time	Minutes
04.00 p.m.	The meeting was called to order.
04.10 p.m.	Every team member described their work on the relevant topics.
04.45 p.m.	Ideas were pitched in by all the group members to make the report more creative.
05.15 p.m.	A deadline was set for the team members to do more research on their relevant topics and create notes and points for the presentation.
05.30 p.m.	The meeting was adjourned.

3rd Coursework Meeting

Date: 12th of November 2022

Time: 09.00 a.m. – 11.00 a.m.

Venue: Virtual Zoom meeting

Attended by:

1) M. G. B. Y. Dias
2) G. S. M. K. Ganegoda
3) H. M. T. Lehan
4) T. A. Karunarathne
5) A. W. D. D. Abeyrathna
6) J. Asfack Ahamath
7) M. S. M. Aflal

Time	Minutes
09.00 a.m.	The meeting was called to order.
09.15 a.m.	New ideas were pitched by the team members and some changes were made to the report.
09.45 a.m.	The points and notes made by the team members were presented and reviewed by fellow members.
10.30 a.m.	The matters regarding the presentation were discussed.
11.00 a.m.	The meeting was adjourned.

4th Coursework Meeting

Date: 17th of November 2022

Time: 05.00 p.m. – 08.00 p.m.

Venue: Virtual Zoom meeting

Attended by:

1) M. G. B. Y. Dias
2) G. S. M. K. Ganegoda
3) H. M. T. Lehan
4) T. A. Karunarathne
5) A. W. D. D. Abeyrathna
6) J. Asfack Ahamath
7) M. S. M. Aflal

Time	Minutes
05.00 p.m.	The meeting was called to order.
05.10 p.m.	Improvements to the PowerPoint presentation points were made.
05.30 p.m.	Joined another meeting with some specialist in textile industry.
07.00 p.m.	After the meeting, discussed about the facts and added them to the report.
07.30 p.m.	The report was almost completed and shared with all the members of the team
08.00 p.m.	The meeting was adjourned.

5th Coursework Meeting

Date: 23rd of November 2022

Time: 05.00 p.m. – 07.00 p.m.

Venue: Virtual Zoom meeting

Attended by:

1) M. G. B. Y. Dias
2) G. S. M. K. Ganegoda
3) H. M. T. Lehan
4) T. A. Karunarathne
5) A. W. D. D. Abeyrathna
6) J. Asfack Ahamath
7) M. S. M. Aflal

Time	Minutes
05.00 p.m.	The meeting was called to order.
05.15 p.m.	Improvements were made to the report and PowerPoint presentation, and it was finalized.
05.45 p.m.	All the teammates were assigned roles to play during the presentation.
06.30 p.m.	A rehearsal was set to happen on the 28th of November 2022.
06.45 p.m.	Time was fixed for submission of the report and the PowerPoint presentation.
07.00 p.m.	The meeting was adjourned.